



DevOps

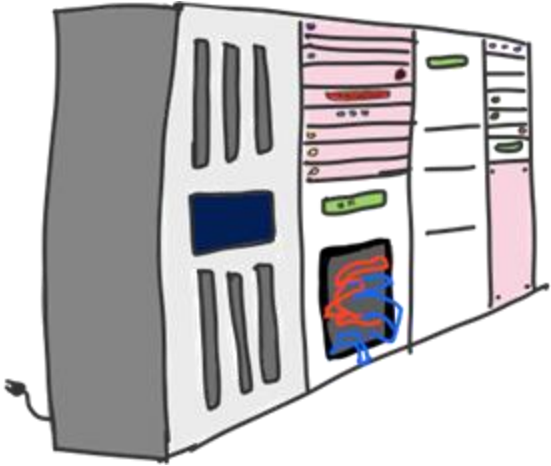
Dr. John-Paul Ore

“Configuration Is Code”

Housekeeping

- Workshop 3 due tomorrow (11:59pm AOE).
- Workshop 1 grades have been released.
- Need clarifications regarding Workshops? use EdStem!

Think-Pair-Share (TPS) Activity:



Docker

- *What are some different ways to specify and create the same computing environment? What if you had to do it 100s of times? What are the issues?*

Today's Learning Objectives

Configuration Management:

*“A process in which an inventory of software and production assets is provisioned and controlled through package managers and tools such as **Ansible**.”*

- Ansible is **Agentless**
- Ansible is **Declarative**
- Ansible is **Idempotent**
- Learn about “Inventory” also known as Hosts a.k.a Managed nodes



ANSIBLE

Intro to Ansible

Michael DeHaan, (NC State 2001)

https://en.wikipedia.org/wiki/Ansible_(software)

☆

J

≡

WIKIPEDIA
The Free Encyclopedia

Search Wikipedia

Search

Create account Log in ...

Ansible (software)

19 languages

Contents [hide]

(Top)

History

Architecture

Overview

Dependencies

Control node

Design goals

Modules

Inventory configuration

Playbooks

Ansible Automation Platform

Platform support

AnsibleFest

See also

References

External links

Article Talk

Read Edit View history Tools

From Wikipedia, the free encyclopedia

This article is about the software named Ansible. For other uses, see [Ansible \(disambiguation\)](#).



This article **may be too technical for most readers to understand**. Please [help improve it](#) to [make it understandable to non-experts](#), without removing the technical details. *(April 2022)*

[\(Learn how and when to remove this template message\)](#)

Ansible is a suite of software tools that enables [infrastructure as code](#). It is [open-source](#) and the suite includes software [provisioning](#), [configuration management](#), and [application deployment](#) functionality.^[2]

Originally written by Michael DeHaan and acquired by [Red Hat](#) in 2015, Ansible is designed to configure both [Unix-like](#) systems and [Microsoft Windows](#). Ansible is [agentless](#), relying on temporary remote connections via [SSH](#) or [Windows Remote Management](#) which allows [PowerShell](#) execution. The Ansible control node runs on most Unix-like systems that are able to run [Python](#), including Windows with [Windows Subsystem for Linux](#) installed.^[3] [System configuration](#) is defined in part by using its own [declarative language](#).

History

[edit]

The term "[ansible](#)" was coined by [Ursula K. Le Guin](#) in her 1966 novel [Rocannon's World](#),^[4] and refers to fictional instantaneous communication systems.^{[5][6]}

Ansible



ANSIBLE

Original author(s)

Michael DeHaan

Developer(s)

[Ansible Community](#) / [Ansible Inc.](#) / [Red Hat Inc.](#)

Initial release

February 20, 2012; 11 years ago^[citation needed]

Stable release

2.15.2^[1] / 18 July 2023

Repository

[github.com/ansible](#) / [ansible](#)

6



Trophy Brewing & Taproom
Craft Beer
+ Community

Dorothea Dix Park

Vault Craft Beer

BOYLAN HEIGHTS

Pullen Park

Residence Inn by Marriott Raleigh...
4.6 (852)
3-star hotel

Heritage Park

Raleigh Convention Center

The Pit Authentic Barbecue
Barbecue • \$\$

Raleigh Union Station

Raleigh Marriott City Center
4.4 (2988)
3-star hotel

Red Hat

The Car Park

Sheraton Raleigh
4.0 (2187)
3-star hotel

Wake County EMS

Whiskey Kitchen

Wake County Justice Center

Wake County Sheriff's Office

Downtown Raleigh

8-bit to 5G Mural
by Taylor White and...

Duke Energy

Union Special

The UPS Store

UPService: Online Business Card Design...

Manhattan Cafe

Reliable Jewelry & Loan
Jewelry store

The Edison Lofts
Apartments



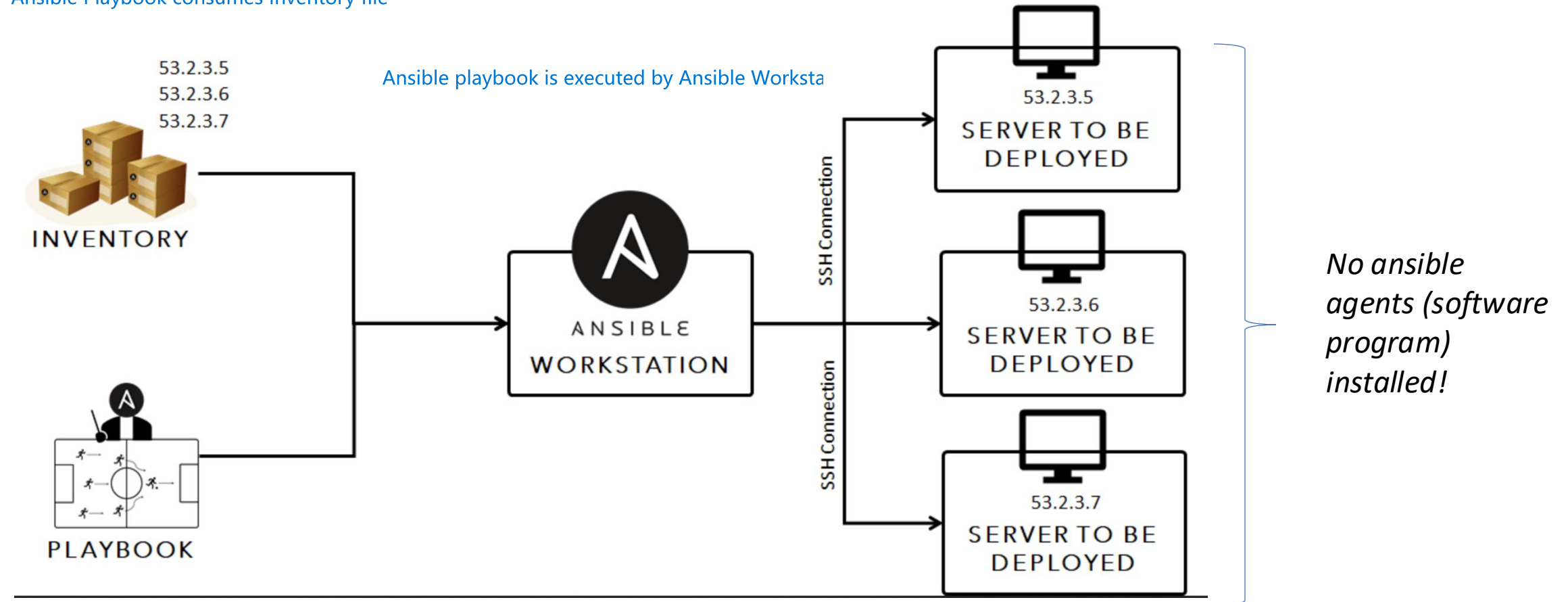
45 minutes.
Recommend the chapters on Ansible, cultivating initial seed users, support, and commercialization.

Ansible at a High-level

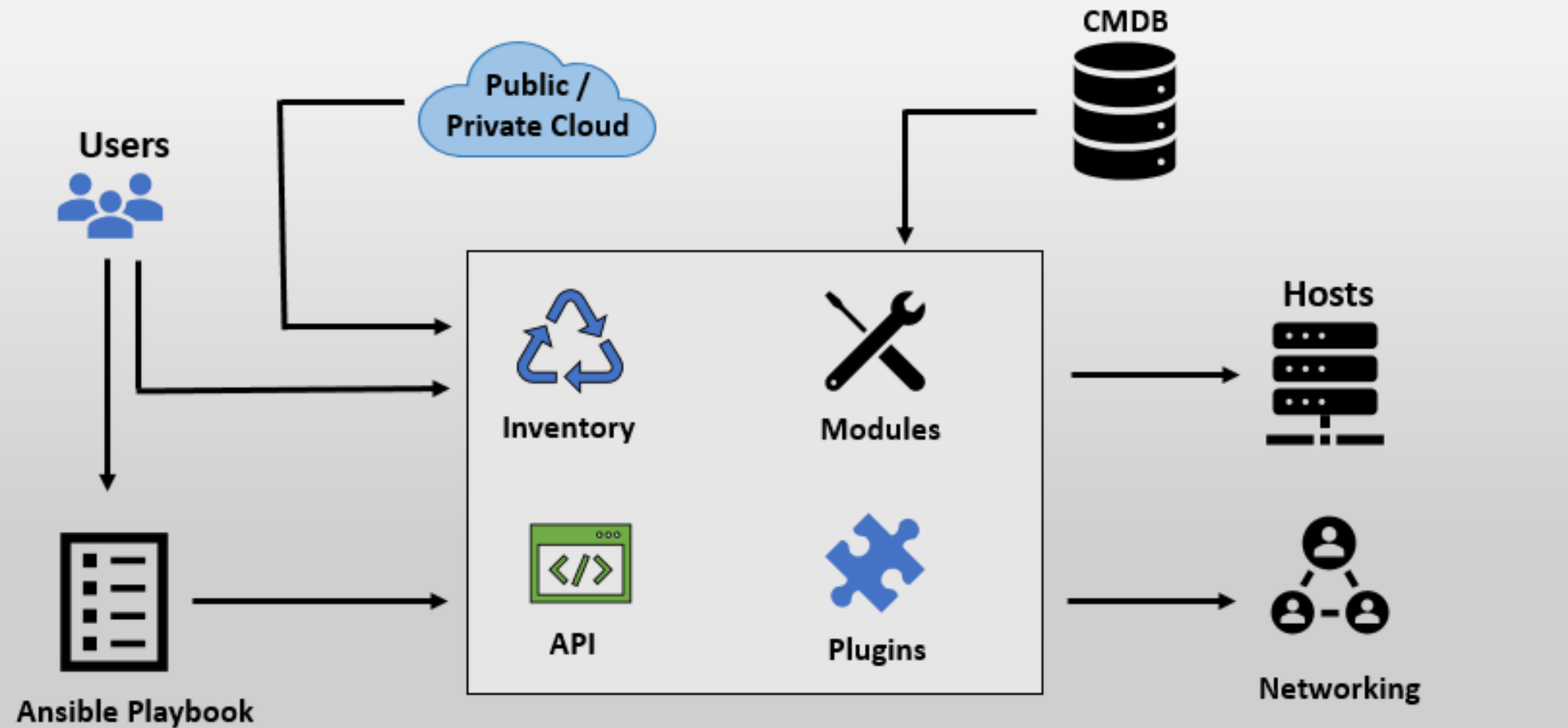
Ansible Workstation is a program that interprets instructions inside the playbook and translated those into instructions on machine.

Agentless - No ansible program needs to be configured on these machines

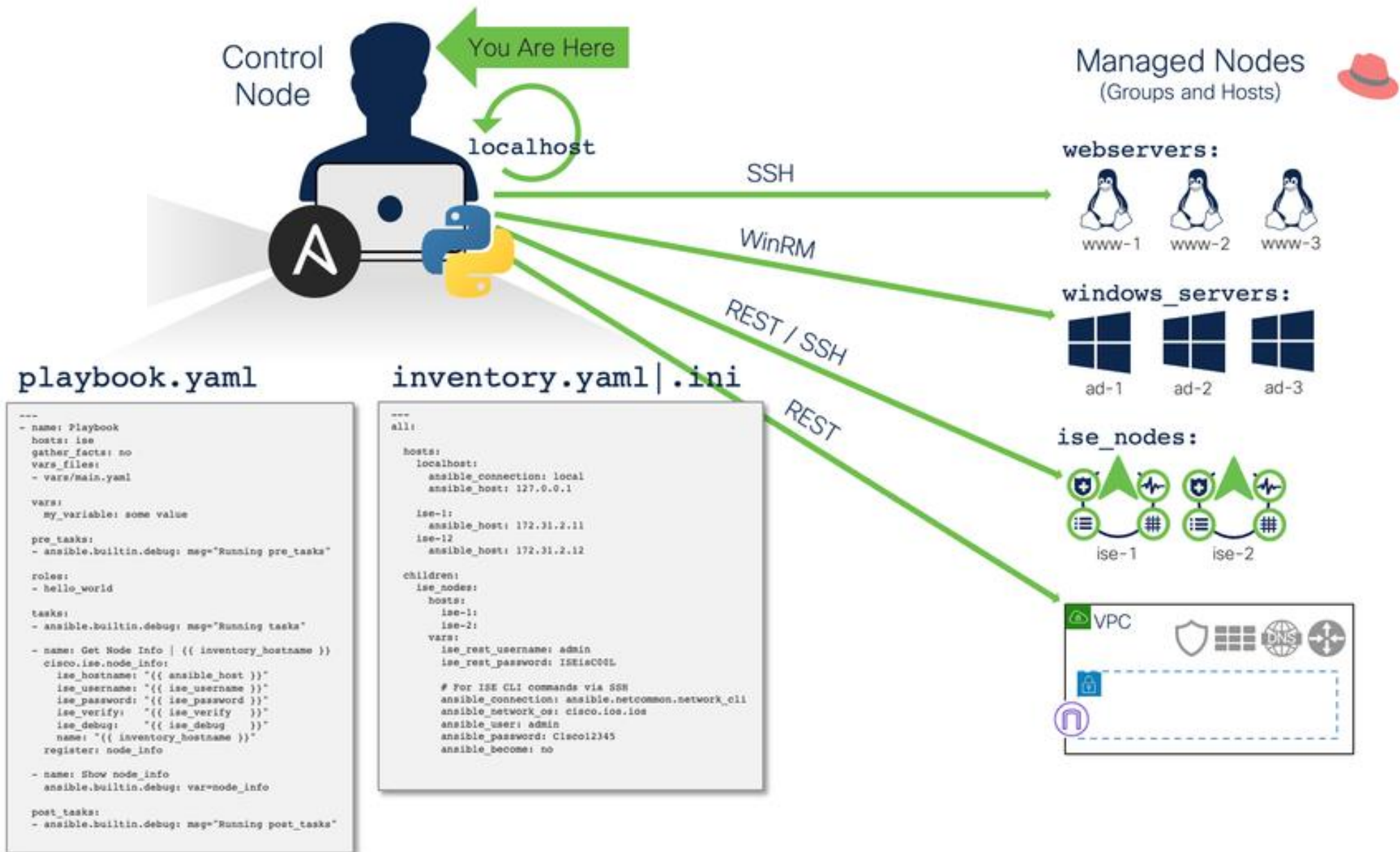
Ansible Playbook consumes Inventory file



Ansible Architecture

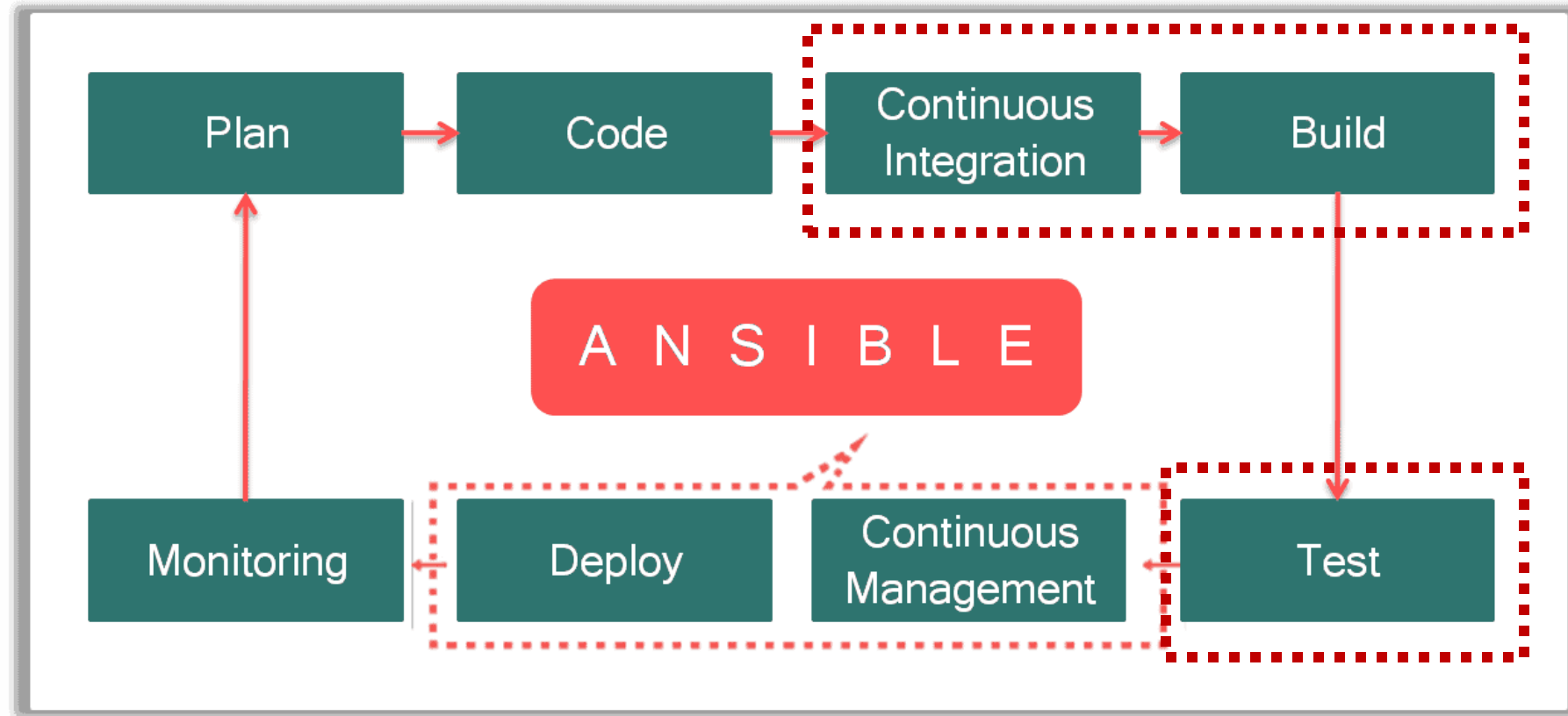


www.educba.com

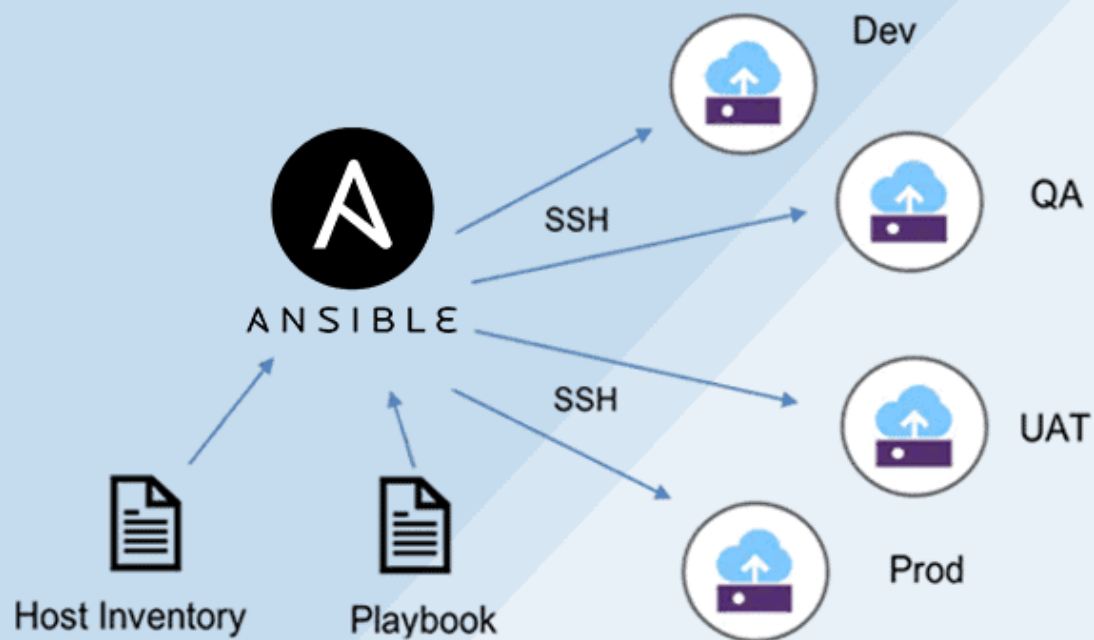


How does Ansible fit into a SE lifecycle?

GitHub Actions can run Ansible Notebook, but not the other way around

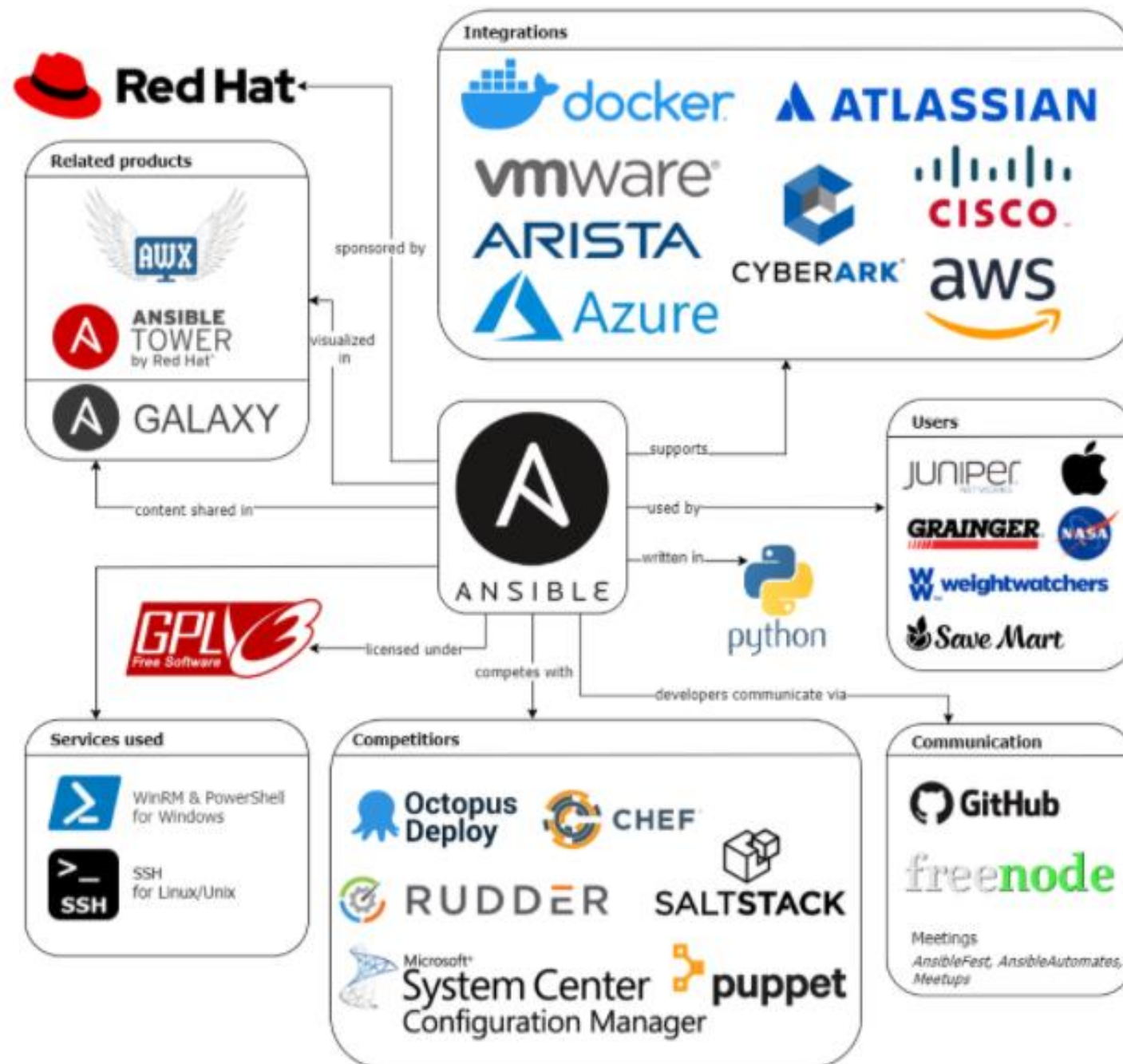


Src: <https://k21academy.com/devops-foundation/ansible-playbook-galaxy-tower/>

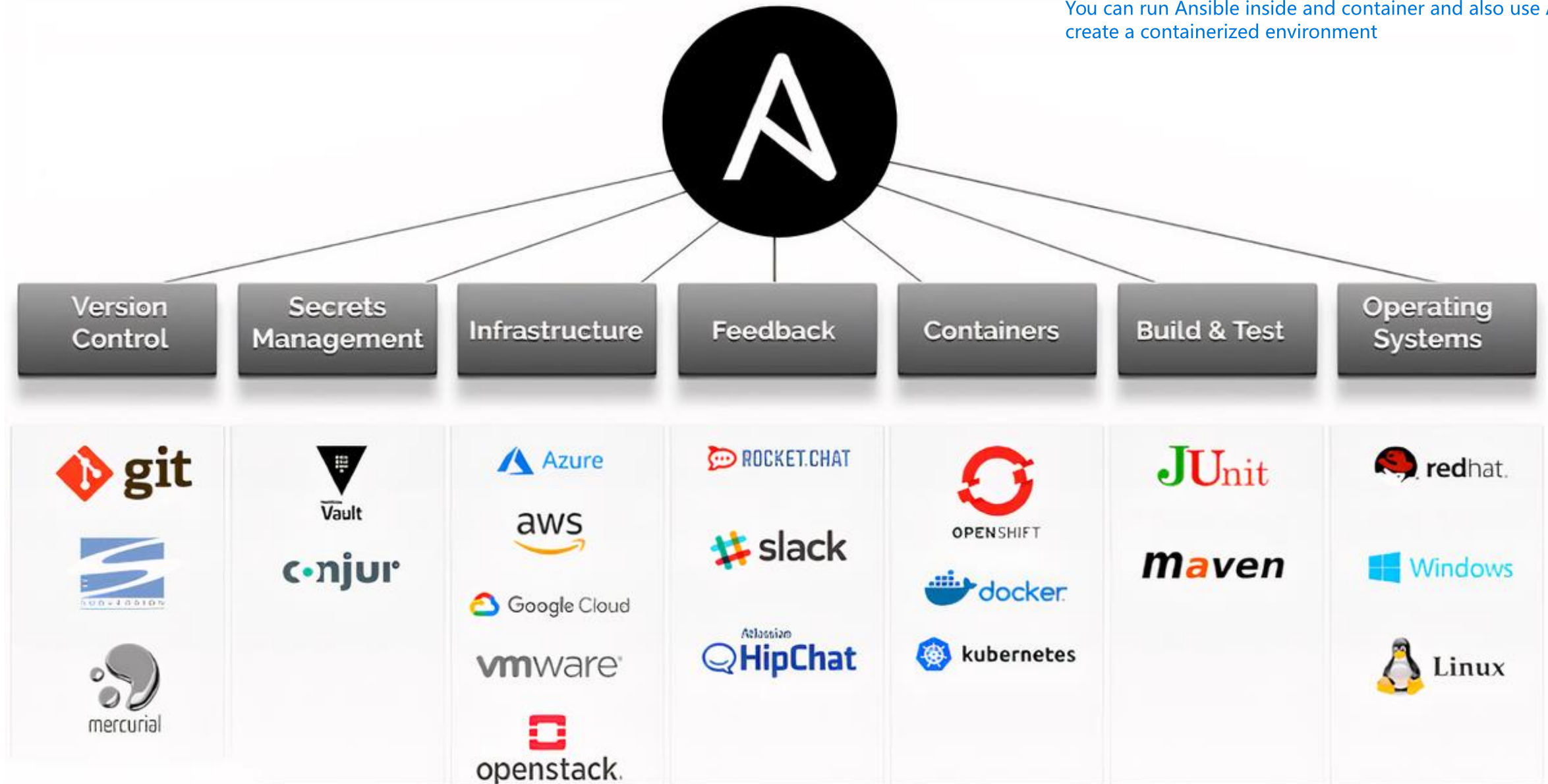


DevOps Foundation

Ansible



You can run Ansible inside and container and also use Ansible create a containerized environment



Ansible is Declarative

Paradigm	Imperative	Declarative
Operation	Defines HOW tasks should be accomplished	Defines WHAT tasks should be accomplished
Computation	Defines the control flow and states changes	Defines only the logic
Mutating Variables	Very usual	Unusual and not recommended
Advantages	Easy to learn notation; Machine architecture compliant	Easy to optimize codes; High abstraction level
Disadvantages	Hard debugging; Vulnerable to data race	Unfamiliar notation; Less customizable codes
Popular Derived Paradigms	Procedural; Object-oriented	Functional; Logic

Src: <https://www.baeldung.com/cs/imperative-vs-declarative-programming>

Control Nodes & Managed Nodes

Control Nodes

- “**Any machine with Ansible installed.** You can run Ansible commands and playbooks by invoking the `ansible` or `ansible-playbook` command from any control node. You can use any computer that has a Python installation as a control node - laptops, shared desktops, and servers can all run Ansible.”

Managed Nodes

“The network devices (and/or servers) you manage with Ansible. Managed nodes are also sometimes called “hosts”. ***Ansible is not installed on managed nodes.***”

a.k.a

“inventory” and “hosts”

Inventory file is also called Ansible Managed Node

Ansible Managed Nodes

Inventory also knows as Hosts a.k.a Managed nodes

“Inventory” in `hosts.yaml`

```
host1.domain.com
```

```
192.168.1.193
```

```
[routers]
```

```
router1.domain.com
```

```
router2.domain.com
```

```
CSR-1 ansible_host=192.168.1.199
```

```
[webservers]
```

```
www1.domain.com
```

```
www2.domain.com
```

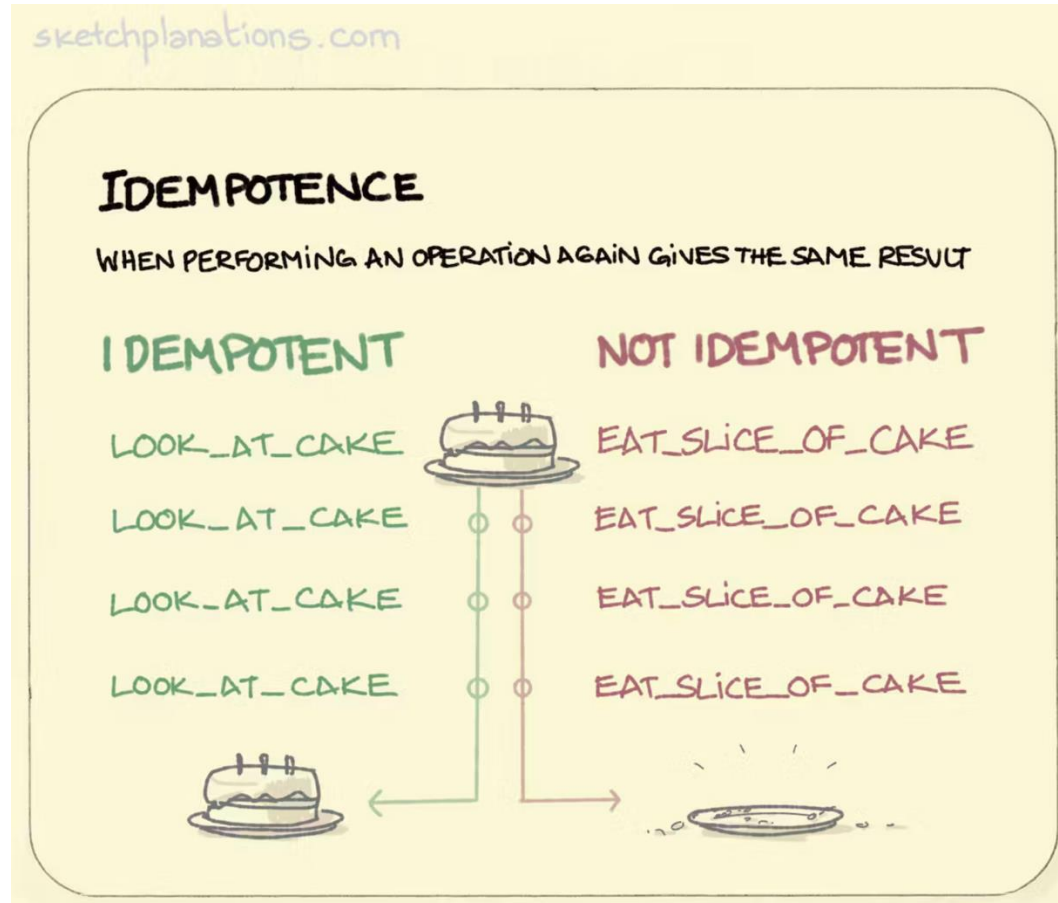

Playbook example #1: `install_webserver.yaml`

```
1  - name: Configure a basic Nginx web server
2    hosts: webservers
3    become: yes
4    tasks:
5      - name: Ensure Nginx is installed (on Debian/Ubuntu)
6        ansible.builtin.apt:
7          name: nginx
8          state: present
9
10     - name: Ensure the Nginx service is running and enabled
11       ansible.builtin.service:
12         name: nginx
13         state: started
14         enabled: yes
```

apt is a
package
manager

Idempotence

state remains the same no matter how many times you look at it



IDEMPOTENCE		
WHEN PERFORMING AN OPERATION AGAIN GIVES THE SAME RESULT		
HTTP METHOD	IDEMPOTENCE	SAFETY
GET	YES	YES
HEAD	YES	YES
PUT	YES	NO
DELETE	YES	NO
POST	NO	NO
PATCH	NO	NO

NORDICAPIS.COM

Playbook example #2

Ansible notebook is idempotent

```
---
- name: Playbook 1 Name of Playbook
  hosts: webservers 2 HostGroup Name
  become: yes 3 Sudo (or) run as different user setting
  become_user: root
  tasks: 4
    - name: ensure apache is at the latest version
      yum:
        name: httpd
        state: latest
    - name: ensure apache is running
      service:
        name: httpd
        state: started
```

Tasks

GitHub Actions & Ansible Playbooks

Actions Only

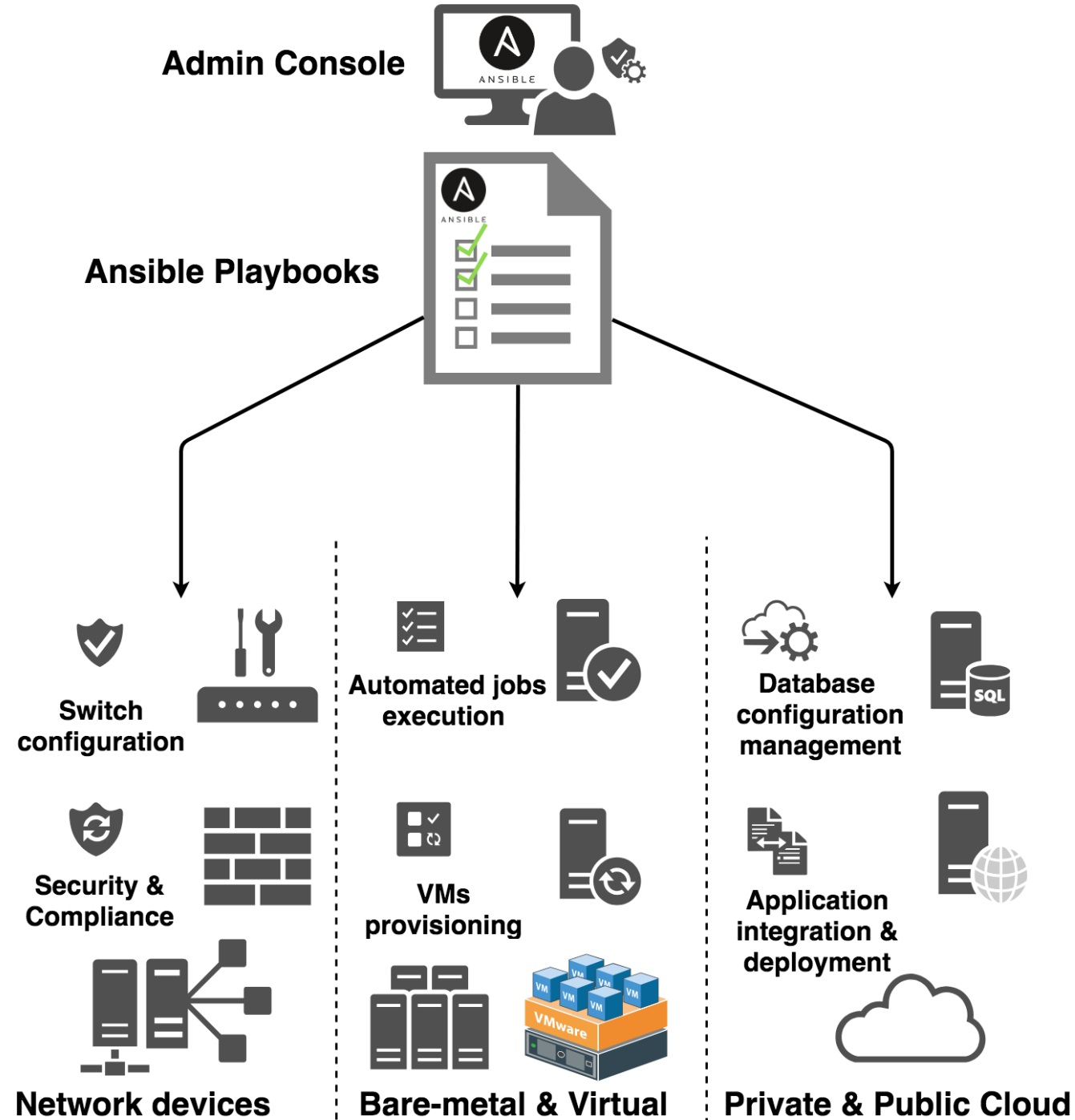
- Not always idempotent because to make GitHub actions idempotent, you have to make extra checks.
- Automates Git repo's lifecycle. Tied to Git Events.
- Runs on a “*runner*” with each jobs in its own clean environment.
- Runs Ansible Playbooks.

Both Actions & Playbooks

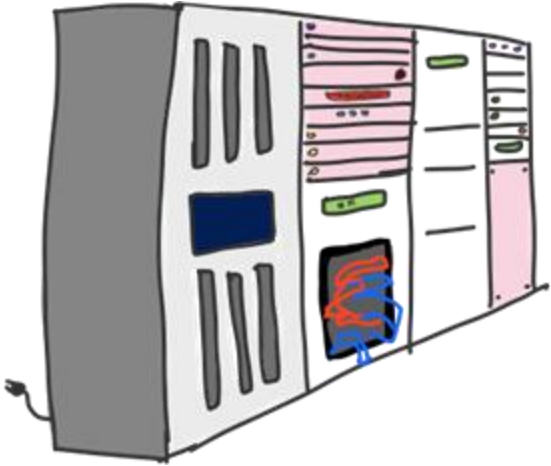
- **Yaml**
- Declarative Syntax
- Used in CI/CD

Playbooks Only

- Always Idempotent.
- Automates Configuration Management and automated Application Deployment.
- Executed from a *control node* that connects to *target nodes*.
- Does not trigger GitHub Actions.



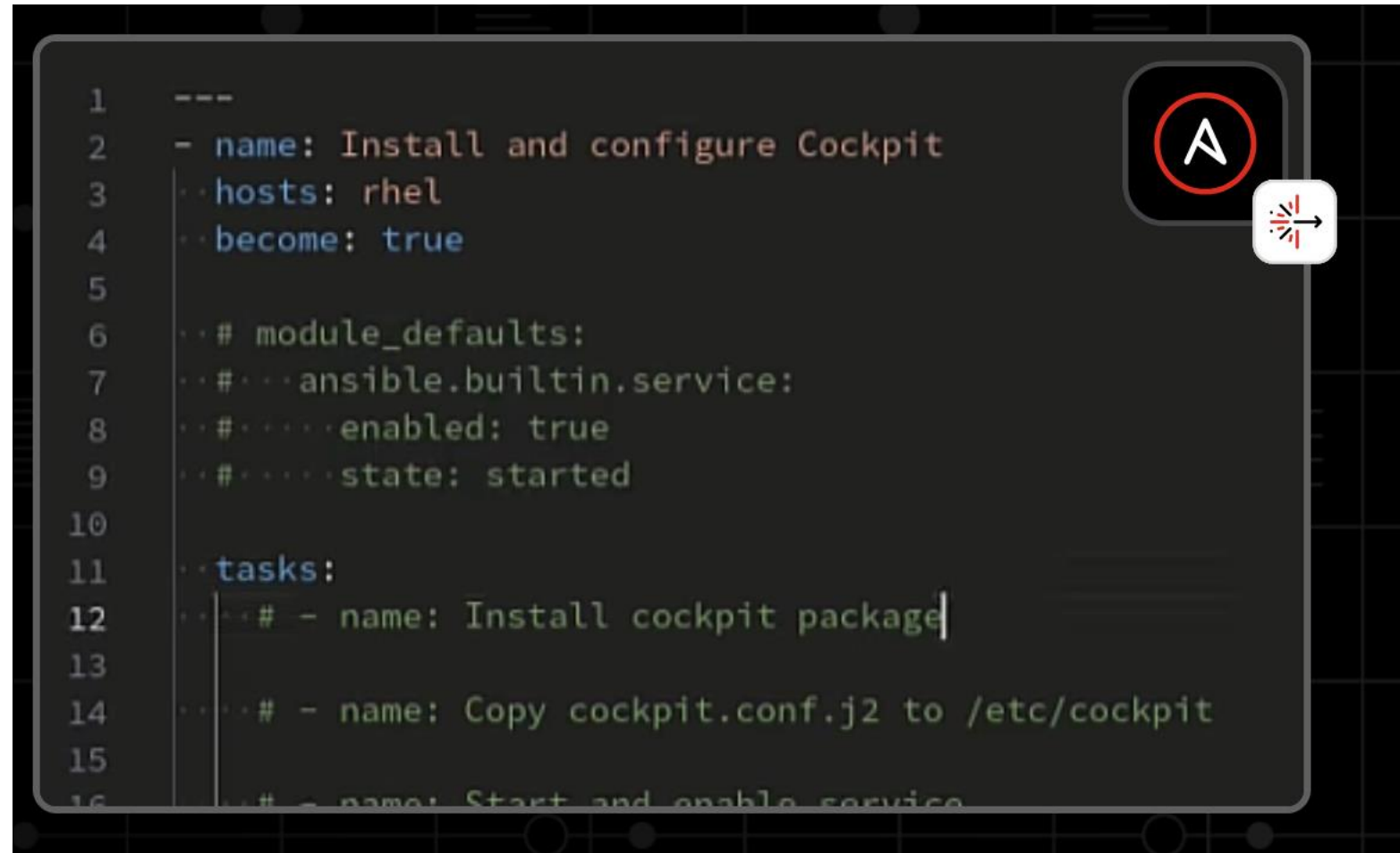
Think-Pair-Share (TPS) Activity:



- *AI Prompts & Ansible: How can I get better results for configuration management tasks?*

Ansible Lightspeed (RedHat AI for Ansible)

- Proprietary.
- Specialist AI
- Comment Completion workflow.



The screenshot displays a dark-themed code editor with a light gray border. On the right side of the editor, there is a red circular icon with a white 'A' and a small white square icon with a red sunburst and an arrow. The code is an Ansible playbook snippet for installing and configuring Cockpit. The text is as follows:

```
1  ---
2  - name: Install and configure Cockpit
3    hosts: rhel
4    become: true
5
6    # module_defaults:
7    # ansible.builtin.service:
8    #   enabled: true
9    #   state: started
10
11   tasks:
12     # - name: Install cockpit package|
13
14     # - name: Copy cockpit.conf.j2 to /etc/cockpit
15
16     # - name: Start and enable service
```

<https://www.redhat.com/en/technologies/management/ansible/ansible-lightspeed#get-started>



Security Smells in Ansible and Chef Scripts: A Replication Study

AKOND RAHMAN, Tennessee Technological University, USA
MD RAYHANUR RAHMAN, CHRIS PARNIN, and LAURIE WILLIAMS,
NC State University, USA

Context: Security smells are recurring coding patterns that are indicative of security weakness and require further inspection. As infrastructure as code (IaC) scripts, such as Ansible and Chef scripts, are used to provision cloud-based servers and systems at scale, security smells in IaC scripts could be used to enable malicious users to exploit vulnerabilities in the provisioned systems. **Goal:** *The goal of this article is to help practitioners avoid insecure coding practices while developing infrastructure as code scripts through an empirical study of security smells in Ansible and Chef scripts.* **Methodology:** We conduct a replication study where we apply qualitative analysis with 1,956 IaC scripts to identify security smells for IaC scripts written in two languages: Ansible and Chef. We construct a static analysis tool called Security Linter for Ansible and Chef scripts (SLAC) to automatically identify security smells in 50,323 scripts collected from 813 open source software repositories. We also submit bug reports for 1,000 randomly selected smell occurrences. **Results:** We identify two

- name: Add nginx repo to yum sources list

yum_repository:

name: "nginx"

file: "nginx"

description: "NGINX repo"

baseurl: "{{ elastic_nginx_repo.repo }}"

state: "{{ elastic_nginx_repo.state }}"

enabled: yes

gpgcheck: no

Disabled 'gpgcheck': no integrity check

Fig. 1. An example Ansible script where integrity check is not specified.

Summary

- Ansible is a configuration management tool.
- Ansible is agentless.